Methodological Blueprint

Agricom, ESMT

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Arjun Chahal, Jean-Christophe Scheider, Akos Peter,

Xishuang Wang, Tsung-Han Tsai, Matias Cam

Table of Contents:

1.0 Problem Framing & Objectives....3
1.1 Business Problem.... 3
1.2 Primary Research Question.... 4
1.3 Supporting Research Questions....4
1.4 Scope.... 4
2.0 Definitions....5
2.1 What We Are Measuring?.... 5
2.2 Unit of Analysis....5
2.3 Granularity....6
2.4 Demand Proxy (Dependent Variable)....6
2.5 Aggregation Logic....7
2.6 Future Target Variables - Dependent (for Real Data Application).... 7
2.7 Independent Variables (Signal List).... 8
2.8 Interaction Effects (Hypothesized)....9
2.9 Leading Indicators.... 9
3.0 Variable Architecture & Hypotheses.... 10
4.0 Data Requirements & Sources....11
4.1 Internal Data Sources Requested:.... 11
4.2 External Data Sources Collected & Reasoning:.... 11
4.3 Other External Data Sources Explored & Limitations:.... 12
4.4 Insight Extraction & Feature Measurement:.... 12
5.0 Data Quality & Handling.... 13
5.1 Missing Value Management.... 13
5.2 Outliers, Scaling, & Transformation Logic.... 14
5.3 Consistency & Alignment Checks....15
5.4 Exploratory Data Analysis (EDA)....16
5.5 Feature Engineering....16
5.6 Noise Filtering & Robustness....18
6.0 Validation Plan.... 19
6.1 Model Utility & Baseline-First Approach.... 19
6.2 Model Selection.... 19
6.3 Safeguarding Against Invalid "Accuracy" Claims.... 20
6.4 Next Steps....20
7.0 Literature Anchor....21
Executive Summary....22
References....23
Appendix....26

1.0 Problem Framing & Objectives

1.1 Business Problem

Agricom is an AI-driven agricultural planning platform operating in the (Berlin) organic produce market. Like many players in the fresh produce supply chain, the company faces a fundamental challenge: Demand for organic produce is inherently volatile and difficult to anticipate. Traditional approaches to inventory planning rely on historical sales data, which is backward-looking by nature. By the time a demand shift shows up in sales figures, the opportunity to act on it has already passed. This reactive approach leads to two costly outcomes: overordering, which drives up spoilage and waste, and underordering, which means lost revenue and unmet customer demand.

The problem is enhanced by the fact that organic produce demand in Berlin is shaped by a diverse and interacting set of forces. Consumer purchasing behavior shifts with the weather, spikes around holidays and cultural events, responds to media coverage of food sustainability, and varies dramatically across neighborhoods with different demographic profiles. No single signal captures the full picture, and the interplay between these factors makes simple heuristics unreliable.

Agricom needs a forward-looking, data-driven methodology that can translate early external signals into actionable demand forecasts with enough lead time (ideally two to four weeks) in order to inform supply chain decisions before demand materializes.

1.2 Primary Research Question

Can publicly available alternative data signals — such as search trends, weather patterns, news sentiment, event schedules, and economic indicators — be used to forecast organic produce demand in Berlin with sufficient accuracy and lead time to support operational planning?

1.3 Supporting Research Questions

1. Which external signals carry the strongest predictive power for organic produce demand, and over what time horizons do they operate?

2. How does organic produce demand vary across Berlin's neighborhoods, and what drives those differences?

3. What is the relationship between media (and social media) sentiment around organic food and consumer purchasing behavior?

4. To what extent do seasonal events, holidays, and local cultural activities create predictable demand patterns?

5. Can this forecasting methodology be generalized beyond a single market?

While the immediate focus is Berlin, Agricom operates with broader ambitions. The project therefore considers whether the methodology (ie. signal selection, data pipeline, and modeling approach) is replicable and scalable to other cities and product categories.

1.4 Scope

The project is structured in two phases:

- - Phase 1 (Ideation and Signal Evaluation): Brainstorming and evaluating candidate data signals informed by Berlin market segmentation and a review of relevant literature.
- - Phase 2 (Signal Validation): Collecting data for the selected signals, building a forecasting model, and validating whether these signals carry genuine predictive value against a demand proxy.

The geographic focus is Berlin, Germany. The product category is organic produce broadly, with the demand proxy constructed from a combination of search interest, weather conditions, event calendars, news sentiment, and economic indicators. The forecast horizon targets two to four weeks ahead, aligned with the lead times required for supply chain adjustments in fresh produce logistics.

2.0 Definitions

2.1 What We Are Measuring?

We are measuring **organic produce demand at the city/neighborhood level in Berlin**, 2–4 weeks ahead, using alternative external signals (weather, events, news sentiment, search trends, economic indicators) as leading indicators.

2.2 Unit of Analysis

Dimension	Definition
Product	Organic produce category (aggregate, not SKU-level)

Time	Daily granularity (aggregatable to weekly)
Location	Berlin neighborhoods (5 customer profiles)

2.3 Granularity

The level of granularity was chosen for the following reasons:

- **Product:** SKU-level forecasting requires product-level sales data which is unavailable. We aggregate to **category level** (e.g., "organic vegetables") to build a robust signal first.
- **Time: Daily granularity** captures short-term fluctuations (weather, events) while being easily aggregatable to weekly for business planning.
- **Location:** Berlin neighborhoods have distinct demographic and behavioral profiles:
    - **Kreuzberg:** farmers markets, sustainability focus
    - **Mitte:** convenience, premium quality
    - **Charlottenburg:** family health, weekend shopping

2.4 Demand Proxy (Dependent Variable)

Since the project uses alternative data to predict organic produce demand and no direct sales data is available, a synthetic demand proxy is constructed as the target variable for model training.

Component	Formula	Weight
Baseline	Constant value	50

Temperature effect	-0.5 * abs(temperature_2m_mean - 15)	Peaks at 15°C, drops for extreme temps
Weekend bonus	+10 if weekend	+10
Christmas season bonus	+25 if Christmas season	+25
Sentiment effect	sentiment * 5	Positive sentiment increases demand
Event effect	event_count * 2	More events increase demand
Noise	Normal(0, 5), seed=42	Gaussian noise for realism

The final demand_proxy is clipped to a minimum of 10.

2.5 Aggregation Logic

Multiple Linear Regression:

$$\hat{Y} = \beta_0 + \beta_1 \times Index_{Baseline} + \beta_2 \times Index_{Weather} + \beta_3 \times Index^2_{Weather} + \beta_4 \times Index_{Weekend}$$

$$+ \beta_5 \times Index_{Holiday} + \beta_6 \times (Index_{Weekend} \times Index_{Holiday}) + \beta_7 \times Index_{Sentiment} + \epsilon$$

Where y_hat represents the final fitted index value for predicted demand.

2.6 Future Target Variables - Dependent (for Real Data Application)

Once Agricom shares sales data (even anonymized), we will re-calibrate the index using actual sales as the true dependent variable. The current proxy is a **demonstration methodology** that proves the signal-based approach works.

Variable	Type	Description
Quantity Sold	Numeric	Volume of product sold (e.g., units or kilograms) per day or week.
Revenue	Numeric	The total monetary value of sales (Optional; for calculating value-weighted demand).

2.7 Independent Variables (Signal List)

Signal	Source	Hypothesized Direction	Lag Assumption	Feature Importance
Day of week	Engineered	Weekend ↑ demand	Same day	24.1%
Christmas/holiday season	Events cal…	Seasonal ↑ 2–3x	0–7 days	21.5%
News sentiment (organic food)	GDELT	Positive tone ↑ demand	3–7 days	13.1%
Temperature (mean)	Open-Meteo	Moderate temp (15–20°C) ↑ fresh produce	Same day	8.8%
Weekend flag	Engineered	↑ demand Sat–Sun	Same day	7.7%
Precipitation	Open-Meteo	Rain ↓ foot traffic → ↓ demand	Same day	~4%
Google Trends (organic keywords)	Google Tr…	Rising searches ↑ demand 2–4 weeks ahead	14–28 days	~6%

Bundesliga home matches	OpenLiga…	↓ organic demand (competing activity)	Same day	~3%
Consumer confidence	OECD/Eu…	High confidence ↑ premium spending	30–60 days	~2%
Food price inflation	Eurostat CPI	High inflation ↓ premium organic spend	30–60 days	~2%

2.8 Interaction Effects (Hypothesized)

Interaction	Expected Effect
Weekend × Good weather	Amplified demand spike (farmers markets, outdoor shopping)
Christmas season × Google Trends spike	Compound demand peak
Rain × Weekday	Strongest demand suppression
High news sentiment × Weekend	Premium purchase behavior reinforcement

2.9 Leading Indicators

Signal	Lead Time	Why It Works
Google Trends	2–4 weeks	People search before they buy
Weather	Same week	Affects shopping behavior immediately
News Sentiment	1–2 weeks	Shapes consumer attitudes

Events	Known ahead	Predictable demand shifts
Economics	1–2 months	Long-term spending patterns

Desired Result: 2–4 weeks of advance notice for inventory planning.

3.0 Variable Architecture & Hypotheses

- **Search Trends (Google Trends):** Search queries act as a strong proxy for historical demand because consumers typically search before they buy. Tracking terms like "Bio Supermarkt" provides a crucial leading indicator capable of predicting shifts 2-4 weeks ahead of lagging sales data.
- **Weather:** Variables like daily temperature, precipitation, and wind speed, are expected to heavily impact both when people shop and what they purchase. For instance, warmer days would likely naturally drive demand for fresh produce, while rain shifts consumer behavior toward convenience stores or delivery services
- **Events:** Bundesliga matches, public holidays, and major festivals (like the Berlin Marathon) cause immediate changes in foot traffic and shopping behavior. These occurrences create highly predictable, localized demand spikes for specific items, such as snacks and beverages
- **News:** Sentiment findings operate with an expected medium lag assumption of up to a couple of weeks, as shifting local narratives gradually influence the collective psychology and spending appetites of residents. We would expect highly positive local news cycles to exert a positive directional effect on premium purchases by boosting overall consumer confidence; whereas, any negative news cycles would likely trigger defensive shopping behavior and a pivot away from organic produce toward those cheaper staples.

- **Economic:** Macroeconomic indicators would be expected to carry a more long-term lag assumption, taking months or quarters to structurally shape baseline consumer willingness to pay. Rising inflation would exhibit a negative directional effect on premium organic spending as consumers shift to cheaper alternatives; whereas, strong consumer confidence can positively drive long-term organic adoption.

4.0 Data Requirements & Sources

4.1 Internal Data Sources Requested:

1. **Historical Demand**: Weekly demand trends and seasonal historical data.
2. **Order Status**: Confirmed orders and pending order quantities to fulfill.

4.2 External Data Sources Collected & Reasoning:

1. **Weather Data:** Using Open-meteo API to get the Berlin weather for the past 3 years regarding wind speed, temperature, precipitation amount since different kinds of weather can cause shifts in behavior.
2. **Search Interest:** Google Trends from 2021-2026 to track relative search interest and topic popularity regarding different products and bio companies because of the potential impact brought by trends and hypes.
3. **News Sentiment:** GDELT Project from 2024-2026 to monitor the general sentiment in Berlin, since news sentiment can also affect the outcome.
4. **Events Calendar:** Using manual collection and the Bundesliga API to extract main events calendar due to the possibility major events, like football matches or national holidays would cause the spike of demand

5. **Economic Indicators:** Using OECD, Eurostat, World Bank APIs to collect indicators, since economic changes and shifts can also affect the outcome.

4.3 Other External Data Sources Explored & Limitations:

1. **Social & Web Listening:** Using Awario to track mentions on social media from X (Twitter), Instagram, Facebook. However, the platforms and APIs require paid subscriptions.
2. **Limitation:** Google Trends data collection done manually.

4.4 Insight Extraction & Feature Measurement:

1. **Weather Data:** Tracking of daily temperatures (minimum, maximum and average), “precipitation amount”, “wind speed”, “season”, “weekday”, “weekend”, “snowfall”, etc. in order to extract better insight in the weather condition within the corresponding time frame.
2. **Search Interest:** Extracting the number of mentions (Counts) regarding bio brands like “rewe bio”, “edeka bio”, “lidl bio”, “bio supermarkt”, “wochenmarkt” along with produce like “bio gemüse”, “bio tomaten”. In addition, we also include the bio/organic related wordings and phrases such as "alnatura", "bio lebensmittel" or even recipes mentioned like “salat rezepte” and “suppe rezepte” to keep track of the exposure regarding different aspects such as organic brands and produce.
3. **News Sentiment:** Monitoring of articles that mentioned "Bio-Lebensmittel", "organic food", "Berlin" and extracting the tone score to trace the sentiment.
4. **Events Calendar:** Tracking major events that took place in Berlin regarding the date, expected attendance.etc.

5. **Economic Indicators:** Extraction of inflation rate, GDP growth, food price index.etc to grasp the economic situation.

5.0 Data Quality & Handling

5.1 Missing Value Management

Weather: For any missing temperature or precipitation records lasting three days or fewer, the system calculates the missing values based on the surrounding days (linear interpolation). If the gap exceeds three days, the row is flagged rather than estimated to prevent the introduction of “synthetic bias”, which can mask real shifts in the data (Borrohou et al, 2025).

Google Trends provides data at a native weekly frequency. In order to align this with the daily weather data, the weekly value is "forward filled" (carried over) to each day of that week, and these imputed rows are marked with a flag.

News: In periods where there are no news articles being recorded by GDELT, the sentiment score (Average Tone) is set to 0, to represent a neutral stance, and the rows are flagged. This is done in order to prevent the model from assuming that no news means bad news.

As we initially discussed, a Variational Autoencoder (VAE) can treat missing values as latent variables. Instead of a single guess, it can provide a probability distribution for the missing data, allowing the “Demand Forecaster” to actually understand the uncertainty of the data for that specific day.

5.2 Outliers, Scaling, & Transformation Logic

Outliers are flagged but not removed by default so as to avoid data loss. Project training currently uses StandardScaler, which centers the data around a zero mean and unit variance. However, Borrohou et al (2025) notes that standardization is sensitive to outliers. This issue could arise in the case of the large search spikes observed for the supermarkets during Christmas. Thus, the literature would typically recommend a Median-based Robust Scaler to improve resilience against extreme weather events or viral social trends (Borrohou et al, 2025) and ensure that the everyday variations are not compressed into a small range which would reduce the model's sensitivity to any subtle demand shifts.

Variational Autoencoders (VAEs) typically use activation functions in their output layer to reconstruct the original data, which would actually require Min-Max Scaling (Normalization). Normalization is extremely sensitive to outliers and is even less robust to outliers than standardization (Borrohou et al, 2025). In order to mitigate this sensitivity, a two-stage solution can be implemented involving outlier clipping (winsorization) prior to the Min-Max scaling as complementary pre-processing techniques (Boudt et al, 2020). Then, instead of the Z-score thresholds, the VAE identifies outliers by their Reconstruction Error. If the model cannot accurately recreate an input, it is flagged as an anomaly, a technique increasingly used for unsupervised anomaly detection in complex networks (Berahmand et al, 2024). For a certain search spike that occurs alongside a heatwave that has never been seen in the historical training set, the VAE will struggle to recreate that row and the resulting high reconstruction error flags this as an outlier (Berahmand et al, 2024). A Regime Shift is then marked by a sustained increase in average reconstruction error over time.

In regard to the activation functions, a Mixed Output Layer would be used to incorporate both Sigmoid as well as Tanh for reconstruction (Berahmand et al, 2024). The Tanh activation function will be used to effectively capture sentiment and anomalies, by mapping inputs to the [-1, 1] range, it is ensured that the value 0 retains its semantic meaning as neutral. Sigmoid will then be reserved for the binary flags (ex: has_holiday, is_rainy) given its [0, 1] range.

In order to then train the VAE to produce these stable latent signals, two different mathematical lenses would be used to evaluate its reconstruction. The Mean Squared Error (MSE) heavily penalizes large errors because it squares the difference between the real value and the reconstruction, which then ensures that the VAE is highly sensitive to getting the tone of media sentiment or the severity of a weather anomaly correct (Berahmand et al, 2024). The second is Binary Cross-Entropy (BCE), which is "employed when the input data is binary (0 or 1), this function measures the difference between predicted and actual output in terms of binary cross-entropy loss" (Berahmand et al, 2024). The final piece of the combined loss to be considered is the Kullback-Leibler (KL) Divergence, which helps to regulate the latent space to support the stability of the signals we pass to our model (Berahmand et al, 2024).

5.3 Consistency & Alignment Checks

A primary challenge is aligning data integration from diverse sources containing disparate schemas as well as frequencies. As such, temporal alignment of the sub-daily (GDELT), daily (Weather/Events), and monthly (Economic) data is performed to achieve a common weekly frequency in accordance with the Google Trends inputs. This includes specific validation logic to ensure the weekly resampling of weather data, matches the native granularity of Google Trends exactly.

5.4 Exploratory Data Analysis (EDA)

Existing Visualizations:

- Time-Series Decomposition: Visualizing Berlin supermarket searches to identify clear Christmas spikes and seasonal patterns.
- Correlation Matrices: Scatter plots comparing the mean temperature with the search interest to identify the weak negative correlation between temperature and indoor shopping intent.

A key future implementation to ensure data quality would be to set up some automated reporting (example: density plots for outlier visualization), in order to adequately assess the health of the incoming data.

5.5 Feature Engineering

Time Based Features:

Feature	Dimension
day_of_week	0 (Monday) to 6 (Sunday) from the date
day_of_month	1--31 from the date
week_of_year	ISO week number (1--53)
month	Calendar month (1--12)
quarter	Calendar quarter (1--4)
year	Calendar year

is_weekend	1 if Saturday or Sunday, else 0

Seasonal Features:

Feature	Dimension
is_winter	1 if month in {12, 1, 2}
is_spring	1 if month in {3, 4, 5}
is_summer	1 if month in {6, 7, 8}
is_fall	1 if month in {9, 10, 11}
is_christmas_season	1 if month = 12 and day >= 15
is_easter_season	1 if month in {3, 4} and ISO week in {12, 13, 14, 15}

Weather-Derived Features:

Feature	Logic	Purpose
temp_range	temperature_2m_max - temperature_2m_min	Captures daily temperature variability
is_rainy	precipitation_sum > 1.0 mm	Binary flag for rain days
is_hot	temperature_2m_max > 25°C	Binary flag for hot days
is_cold	temperature_2m_min < 5°C	Binary flag for cold days
is_windy	wind_speed_10m_max > 40	Binary flag for windy days

	km/h	
month	Extracted from date	Calendar month (1--12)
season	Mapped from month	winter (Dec--Feb), spring (Mar--May), summer (Jun--Aug), autumn (Sep--Nov)
day_of_week	Extracted from date	Day of week (0=Monday, 6=Sunday)
is_weekend	day_of_week in [5, 6]	Binary flag for weekends

5.6 Noise Filtering & Robustness

In order to reduce instability in our collected data we apply 7-day and 14-day rolling means to help smooth out high-frequency noise in daily sentiment and weather observations. Furthermore, while encoding categorical variables is necessary, Borrohou et al. (2025) warns it can lead to "feature explosion". Currently this is mitigated by selecting only the top 10 features via Random Forest importance ranking before final ensemble prediction.

6.0 Validation Plan

6.1 Model Utility & Baseline-First Approach

In their survey on Autoencoders, Berahmand et al. (2024) highlight that while autoencoders are powerful for both feature extraction and dimensionality reduction, their primary value is derived from capturing those non-linear relationships that traditional methods will miss. Therefore, by starting with some linear validation with which to baseline, we can define exactly what the

"linear signal" is, making it clear what the VAE is adding in terms of that non-linear "latent" information. Current validation acknowledges the use of a synthetic demand proxy, which introduces circularity. Despite this, the current validation can showcase:

- Feature Dominance: describing which parts of the proxy are the most "stable." For example, if the model ignores weather but clings to sentiment, it showcases that media tone has a more consistent relationship with the proxy.
- Predictive Precedence: Granger Causality tests can describe if if signal X "granger-causes" demand Y (the proxy in this case), then past values of X should contain information that helps predict Y above and beyond what can be predicted using only past values of Y (Sharma et al, 2022)

6.2 Model Selection

Stress tests: examining whether features have predictive value in an interpretable, linear context before moving to a probabilistic, latent context.

Model Category	Potential Candidates	Our Choice & Rationale
Statistical	ARIMA, Exponential Smoothing	Used primarily to establish the "Seasonal Baseline" from which our external signals deviate.
Linear / Regularized	OLS, Ridge Regression	Primary Choice: We selected Ridge Regression because it effectively handles multicollinearity, a frequent issue when multiple Google search terms (like "organic" and "bio") are highly correlated.

Machine Learning	Random Forest, XGBoost	Used to capture non-linear Interaction Effects, such as the compound effect of "Good Weather × Weekend" on farmers' market demand.

To best mitigate multicollinearity, Ridge Regression applies L2 regularization, which "shrinks" the coefficients of correlated features. The goal is to help establish a stable ranking of which external signals actually have the strongest linear relationship with the demand proxy. This allows us to validate if the previously mentioned "common sense" assumptions (ex. rainy days = less fresh produce demand) are actually supported by the data. Additionally, by extracting feature importance from a Random Forest model, we can validate if the interaction between signals is real. For example, if the model places high importance on the interaction between is_weekend and temp_anomaly, it clearly demonstrates that our data does actually contain complex patterns that a more simple seasonal average would miss out on.

6.3 Safeguarding Against Invalid "Accuracy" Claims

Providing a credible methodology:

- Strict Temporal Validation: We do not use random data shuffling. To prevent "future leakage," we use Time-Series Cross-Validation (Rolling Window), ensuring that the model actually only trains on data that would have been available before the forecast date.

6.4 Next Steps

The current setup examines data collection, pre-processing, feature engineering, aggregation and transformation logic, data quality, and initial output validation. The VAE allows us to validate the input quality through latent space analysis, which represents a shift from reactive error-checking to proactive stability monitoring.

Some future of work steps following alignment with the Agricom team:

- Implementation of two stage pipeline to meet requirements of activation functions & ensure outlier resilience
    - Application of further winsorization based on the interquartile range (IQR).

    - Transition from Standard Scaler to Min-Max
- Implementation of VAE with mixed layer output (Sigmoid, Tanh)
- Validation with Agricom POS data (anonymized) or sample
    - Validate the correlation between search volume and actual sales
    - Re-weight keywords based on the true product mix
    - Build SKU-level models instead of category-level
- Geographic Granularity & Segmentation
- Exploration of the translation of the VAE's output (raw latent vector, etc.) to understandable insights (natural language report on Regime shift, etc.).

7.0 Literature Anchor

Following a review of available literature, it is clear that several findings validate the use of alternative data sources to improve forecasting accuracy. Published research across a variety of industries reflects that these other data signals can effectively capture real-time consumer sentiment and behavior that is often missed by traditional sources, providing leading indicators for forecasting.

Some key takeaways include:

- Cui et al. (2018) found that social media information improved forecast accuracy by 12.85–23.23%, with random forests outperforming linear models.
- Kulkarni et al. (2010) discovered that online search activity follows “rather predictable patterns” and provides valuable measures of consumer interest, demonstrating systematic relationships between search volume and outcomes.

- Tonya Boone et al. (2018) demonstrated that Google Trends customer search data reduces out-of-sample forecast errors. The researchers worked with an online specialty food retailer, analyzing five stock keeping units (SKUs) with at least four years of sales and pricing data. They used Google Trends search index data for specific search terms most likely to lead customers to each product, as identified by the business owner. Across the five SKUs that were studied, incorporating Google Trends data reduced out-of-sample mean absolute percentage errors (MAPE) by 2.2% to 7.7%.
- Fleder et al. (2019) developed approaches for handling extremely “sparse ground-truth” data (like quarterly reports) combined with noisy daily proxy signals.

Executive Summary

Agricom faces an operational challenge: demand volatility cannot be anticipated using backward-looking sales data alone, resulting in costly overordering that drives spoilage and underordering that results in lost revenue. To address this, the project investigates whether publicly available alternative data signals (weather conditions, Google search trends, news sentiment, event calendars, and macroeconomic indicators) can help to forecast organic produce demand across five Berlin neighborhood customer segments with a two-to-four week lead time that is sufficient for proactive supply chain decisions. In the absence of proprietary sales data, a synthetic demand proxy is constructed as the model training target, incorporating temperature, media sentiment, event density, weekend effects, and seasonal factors. The modelling framework adopts a baseline-first approach for interpretable linear signal validation and identification of non-linear interaction effects such as the compound demand impact of good weather coinciding with a weekend. The integration of a Variational Autoencoder is additionally researched to support anomaly detection, missing data imputation, and latent signal stability monitoring. External data is sourced from Open-Meteo, Google Trends, GDELT, the Bundesliga API, and OECD/Eurostat, with rigorous missing value management and flagging protocols in place to preserve data integrity. The methodology is technically sound and well-anchored in published literature; however, its predictive validity is currently constrained by the absence of real sales data. Two decisions are therefore required from Agricom: validation of the current direction and methodology of the project to continue with currently proposed next steps (see section 6.4), and the provision of anonymized point-of-sale data or at minimum a schema and sample rows, which would replace the synthetic proxy with a true dependent variable and unlock the transition to SKU-level modelling.

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Appendix

Open questions:

- Would you prefer the focus city/market to be only on the five segments discussed in **Berlin**, or should we seek to design the framework so it can generalize to additional cities later?
- Even if you prefer not to share full data yet, could you confirm the **data schema** (column names/definitions) or provide a **small anonymized sample** (e.g., a few rows) so we can ensure our assumptions about units and structure are correct?